



Pharmacognostic Characterization of *Kaempferia galanga* L. from Northern Vietnam for Quality Control and Sustainable Medicinal Plant Production

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Abstract: *Kaempferia galanga* L. is a medicinal plant widely used in traditional medicine. However, data on its anatomical and powder microscopic characteristics in Vietnam remain limited. This study aimed to characterize the macromorphological, anatomical, and powder microscopic features of *K. galanga* collected from Bac Ninh Province, Vietnam, and to establish preliminary diagnostic characteristics for botanical authentication and quality control. Twenty fresh plant specimens were collected from spatially separated sampling points and subjected to macromorphological, anatomical, and powder microscopic examinations. The results showed that *K. galanga* is a low-growing herbaceous plant with a characteristic aromatic odor in all examined plant parts. The rhizomes were globose or formed chains of connected globose segments and measured approximately 1.5–2.0 cm in diameter. The leaf sheaths were 3–5 cm long, and the leaf blades were oval to broadly ovate, measuring 15–20 × 10–15 cm. The inflorescences bore 15–20 flowers, each approximately 6 cm long and predominantly white, with a characteristic purple-striped labellum. Anatomically, the leaves exhibited an isobilateral, amphistomatic structure, with vascular bundles distributed within the mesophyll and well-developed aerenchyma. The rhizomes possessed a cork layer comprising 6–15 cell layers, whereas the roots had a cork layer comprising 3–10 cell layers. The rhizomes contained abundant starch granules and oil-containing cells, while oil-containing cells were also observed in the roots. Both underground organs displayed characteristic vascular-tissue arrangements of the family Zingiberaceae. Powder microscopy of the leaves and rhizomes revealed potentially useful diagnostic features, including epidermal fragments bearing stomata (approximately 30 × 50 μm), starch granules (20–30 μm), prismatic calcium oxalate crystals (approximately 20 × 30 μm), spiral vessels, fibers, and cork fragments. These findings provide baseline pharmacognostic information for the authentication and quality assessment of *K. galanga* from northern Vietnam. Once validated, the proposed diagnostic characteristics may support the identification and traceability of authentic planting materials, quality assurance, germplasm conservation, and sustainable medicinal plant production, including production systems incorporating organic principles. Further validation using samples from broader geographical populations and complementary molecular or phytochemical approaches is required.

Keywords: *Kaempferia galanga* L.; Pharmacognosy; Botanical authentication; Powder microscopy; Medicinal plant quality control; Planting-material traceability; Sustainable medicinal plant production

1. Introduction

Kaempferia galanga L. is a perennial herbaceous plant belonging to the family Zingiberaceae (Kumar, 2020; Singh et al., 2023). The species was described by Carl Linnaeus in 1753 (Kumar, 2020). The genus *Kaempferia* comprises approximately 50–60 species distributed mainly across the humid tropical regions of Asia, including Vietnam, Laos, Cambodia, Thailand, Malaysia, Borneo, Sumatra, and India (Kumar, 2020; Silalahi, 2021). In Vietnam, *K. galanga* has been reported in several regions and is cultivated as a medicinal and spice plant (Chuong et al., 2023; Do et al., 2026). However, the classification and accurate identification of species within the genus

Kaempferia remain challenging because of their considerable morphological similarity. In particular, *K. galanga* shares several external characteristics with related species, such as *K. rotunda* and *K. parviflora* (Wongsuwan et al., 2025). Consequently, external morphology alone may be insufficient for reliable species authentication (Labrooy et al., 2018; Wongsuwan et al., 2025).

Accurate species identification is essential for the appropriate utilization and medicinal application of plant resources. Polyploidization, hybridization, genetic variation, and similarities in gross morphology may contribute to the taxonomic complexity of the genus (Basak et al., 2018; Nopporncharoenkul et al., 2017; Wongsuwan et al., 2025). To improve taxonomic accuracy and quality control, studies conducted in several Asian countries have examined the morphological, anatomical, cytological, or molecular characteristics of *K. galanga* and related *Kaempferia* species. Detailed descriptions have been reported from Thailand (Saensouk & Saensouk, 2021; Sirirugsa, 1989; Wongsuwan et al., 2025), India (Gangmei et al., 2024; Uma & Muthukumar, 2014), Malaysia (Labrooy et al., 2018), China (Lu et al., 2023), and Myanmar (Rivera et al., 2026). These studies have provided valuable taxonomic information and demonstrated the importance of integrating anatomical and microscopic characteristics with external morphology for reliable species authentication.

Beyond its taxonomic importance, *K. galanga* has attracted considerable attention because of its pharmacological potential. Essential oils extracted from the rhizomes and leaves exhibit broad-spectrum antimicrobial activity, while the rhizomes contain high levels of ethyl *p*-methoxycinnamate, one of the principal bioactive constituents associated with several reported biological activities of the species (Begum et al., 2023; Wang et al., 2023; Wang et al., 2021). Studies conducted in Vietnam have also identified numerous bioactive compounds and demonstrated the medicinal potential of Vietnamese *K. galanga* materials (Do & Dao, 2026; Men et al., 2020; Men et al., 2024; Men et al., 2022). However, despite these advances, pharmacognostic information on *K. galanga* collected in Vietnam remains limited. To the best of our knowledge, few studies have comprehensively integrated the macromorphological, anatomical, and powder microscopic characteristics of its rhizomes, roots, leaves, and powdered materials. Addressing this knowledge gap may improve species authentication, support pharmacopoeial standard development, and strengthen the quality control of Vietnamese medicinal materials.

Macromorphological description has traditionally served as an important method in plant taxonomy; however, combining it with anatomical and powder microscopic examinations can improve the reliability of plant identification (Alamgir, 2017). Anatomical examination of organs such as rhizomes, roots, and leaves may assist in authenticating plant materials and detecting substitution or contamination (Alamgir, 2017; Osman et al., 2019). Powder microscopy is also useful for identifying dried and powdered crude drugs in which external morphological characteristics are no longer preserved (Osman et al., 2019).

In Vietnam, *K. galanga* is cultivated by households and medicinal plant growers and is used as both a traditional medicinal plant and a spice crop (Chuong et al., 2023). Increasing demand for herbal medicines and natural products has also stimulated interest in improving its cultivation and quality management under good agricultural and collection practices (GACP) and other traceable production systems (Graf vom Hagen-Plettenberg et al., 2012; World Health Organization, 2003). However, limited access to authenticated planting materials and the morphological similarity among related *Kaempferia* species may increase the risk of misidentification and substitution. Reliable botanical authentication is therefore relevant to quality assurance, raw-material traceability, germplasm conservation, and sustainable medicinal plant production. Accordingly, this study aimed to characterize the macromorphological, anatomical, and powder microscopic features of *K. galanga* collected from Bac Ninh Province, Vietnam, and to establish preliminary diagnostic criteria for botanical authentication and quality control. By supporting the identification and traceability of authentic planting materials, the findings may also contribute to quality assurance in sustainable medicinal plant production and production systems incorporating organic principles.

2. Methodology

2.1 Plant Materials

Twenty fresh specimens of *K. galanga* were collected in May 2024 from spatially separated sampling points in Luc Ngan District, Bac Ninh Province, Vietnam (21.383384° N, 106.691027° E), during the active vegetative and flowering period. To reduce the repeated collection of individuals from the same clonal cluster, the selected plants were separated by at least 2–3 m, and only one individual was collected from each sampling point. Plants showing visible symptoms of disease or mechanical damage were excluded. The collected specimens were thoroughly washed with water to remove adhering soil and surface impurities before macromorphological and anatomical examination. Portions of the leaves and rhizomes were air-dried at room temperature, separately ground into powder, and used for powder microscopic examination. The specimens were authenticated on the basis of their macromorphological characteristics using the Vietnamese Pharmacopoeia V (Ministry of Health of Vietnam, 2017) and relevant taxonomic literature on the family Zingiberaceae. A voucher specimen was prepared and deposited

at the Department of Pharmacognosy and Traditional Medicine, Hanoi University of Pharmacy, for future reference (voucher specimen No. 95/2024).

2.2 Macromorphological Examination

Macromorphological examination was conducted according to the procedures described in the Vietnamese Pharmacopoeia V (Ministry of Health of Vietnam, 2017). All available vegetative and reproductive organs of the collected specimens were examined, including the adventitious roots, rhizomes, leaf sheaths, leaf blades, inflorescences, flowers, and their component structures. External characteristics, including shape, color, surface texture, odor, arrangement, and dimensions, were recorded. Larger organs were measured using a ruler, whereas smaller floral structures were measured using a digital caliper. A handheld magnifying lens was used when necessary to examine smaller surface characteristics. Quantitative dimensions were recorded from the available specimens and are reported as the observed ranges. Representative plant organs and diagnostic characteristics were photographed using a Canon EOS 90D digital camera.

2.3 Microscopic Examination

2.3.1 Anatomical examination

Anatomical characteristics were examined using the iodine green–carmine double-staining method described in the Vietnamese Pharmacopoeia V (Ministry of Health of Vietnam, 2017). The procedure was conducted as follows:

1. Representative samples of fresh roots, rhizomes, and leaves were manually cut into transverse sections approximately 10–20 μm thick using a razor blade. For each plant organ, five sections were prepared from each specimen. Only intact sections showing clear tissue organization were selected for staining, microscopic examination, and photomicrography.
2. The selected sections were cleared with 5.0% (w/v) chloramine-T and 50% (v/v) chloral hydrate for 10 min, followed by treatment with 1.0% (w/v) acetic acid for 2 min. The sections were then stained with 0.3% (w/v) iodine green for approximately 5 s and subsequently with 1.0% (w/v) carmine until adequate tissue differentiation was achieved. Double-distilled water was used to remove excess reagents between successive treatment steps.
3. The stained sections were mounted in a glycerin–water mixture (1:1, v/v), covered with coverslips, and examined under a light microscope (HumaScope, Germany) using 4 $\times$, 10 $\times$, and 40 $\times$ objectives. Anatomical observations were repeated across the prepared sections to confirm the consistency of the observed characteristics.
4. Where applicable, microscopic measurements were performed using a calibrated ocular micrometer, and the dimensions were reported as observed ranges. Representative anatomical characteristics were photographed under identical microscopic conditions.

2.3.2 Powder microscopic examination

Following the procedures described in the Vietnamese Pharmacopoeia V (Ministry of Health of Vietnam, 2017), small quantities of the separately prepared leaf and rhizome powders were placed on glass slides and mounted in glycerin–water (1:1, v/v). The preparations were covered with coverslips and examined using the same HumaScope light microscope with 10 $\times$ and 40 $\times$ objectives. For the leaf powder, the examination focused on epidermal fragments bearing stomata, parenchymatous tissue fragments, spiral vessels, starch granules, prismatic calcium oxalate crystals, and multicellular covering trichomes. For the rhizome powder, spiral vessels, fibers, parenchymatous tissue fragments, individual and clustered starch granules, pigmented masses, and cork fragments were recorded. The dimensions of the stomata, starch granules, calcium oxalate crystals, and covering trichomes were determined using a calibrated ocular micrometer. Clearly resolved structures in representative microscopic fields were measured, and the dimensions were reported as observed ranges. Photomicrographs were obtained under consistent microscopic conditions to document representative powder characteristics.

3. Results

3.1 Macromorphological Characteristics

The examined specimens of *K. galanga* were small, low-growing herbaceous plants, and all examined plant parts had a characteristic aromatic odor (Figures 1a and 1b). The rhizomes were tuberous and globose or formed chains of connected globose segments, measuring approximately 1.5–2.0 cm in diameter. Their outer surfaces ranged from pale yellow to dark brown and bore numerous nodes, whereas the cut surfaces were creamy white (Figures 1c and 1d). The plants generally bore two subsessile leaves positioned close to the ground (Figure 1e). The leaf sheaths were thin, 3–5 cm long, and light green, with hairs on the abaxial surface (Figure 1h). The leaf

blades were oval to broadly ovate, measuring approximately 15–20 × 10–15 cm. The adaxial surface was dark green and nearly glabrous, whereas the abaxial surface was light green and covered with yellowish-brown velvety hairs and globose or formed chains of connected globose segments, measuring approximately 1.5–2.0 cm in diameter. Their outer surfaces ranged from pale yellow to dark brown and bore numerous nodes, whereas the cut surfaces were creamy white (Figures 1c and 1d). The plants generally bore two subsessile leaves positioned close to the ground (Figures 1f and 1g). The leaf margin was entire to slightly undulate and often pale yellow; the base was rounded to obtuse; and the apex was acute to shortly acuminate, sometimes with a purplish-brown tip (Figures 1i and 1j). The venation was parallel.

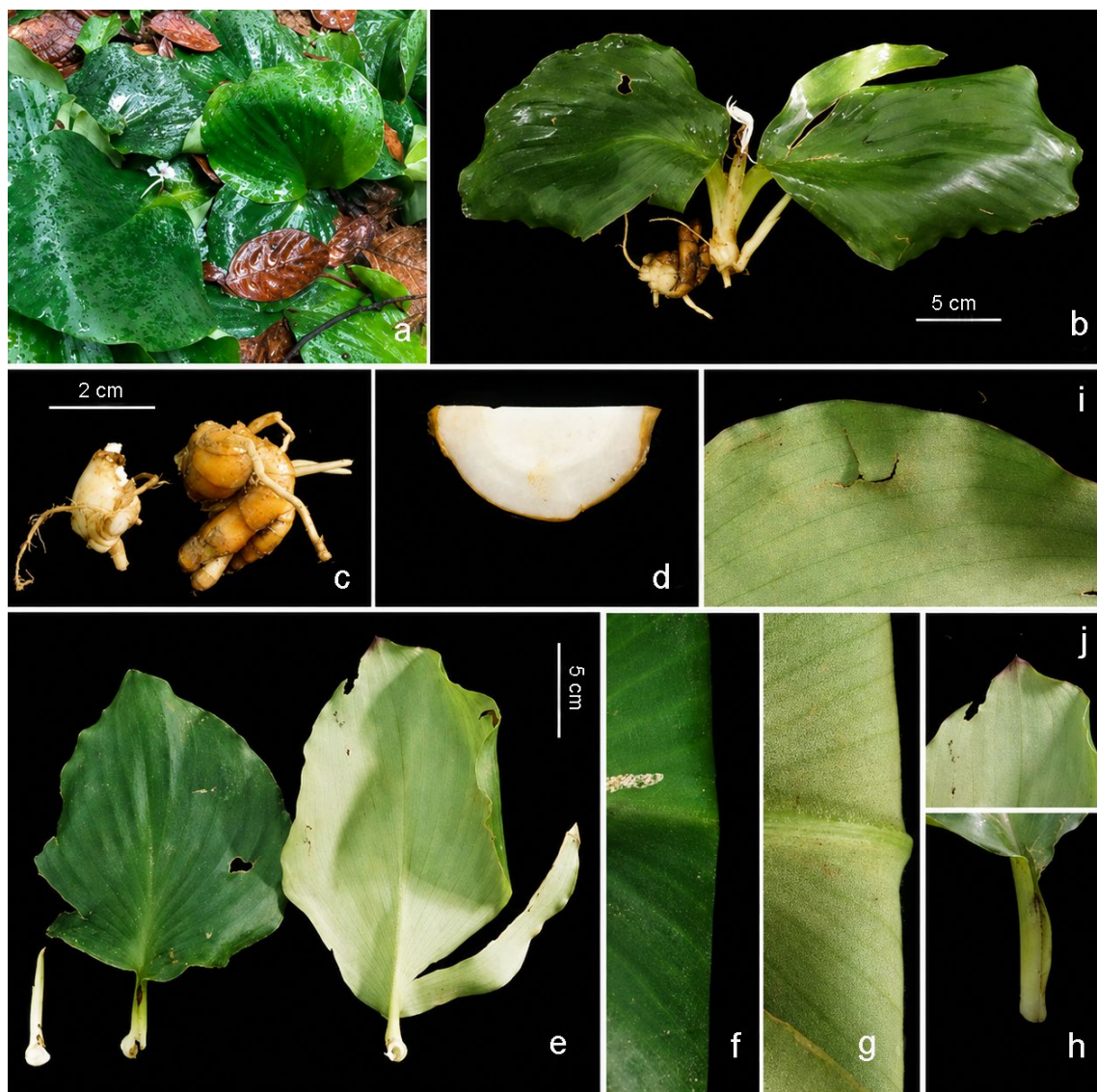


Figure 1. Macromorphological characteristics of the vegetative organs of *Kaempferia galanga* L.: (a) plant in its natural habitat; (b) whole plant; (c) rhizome; (d) transverse section of the rhizome; (e) leaf morphology; (f) adaxial leaf surface; (g) abaxial leaf surface; (h) leaf sheath; (i) leaf margin; (j) leaf apex

The terminal inflorescences emerged between the leaves and consisted of 15–20 flowers enclosed by overlapping sheath-like bracts (Figures 2a–2d). The bracts were boat-shaped, 2–3 cm long and 0.3–0.5 cm wide, greenish-white, thin, and easily wrinkled, with smooth surfaces and acuminate apices (Figures 2c and 2f). The flowers were approximately 6 cm long and predominantly white (Figure 2e). The calyx consisted of three sepals fused at the base and divided apically into two narrow, strap-shaped lobes approximately 3 cm long and 0.2 cm wide. The lobes were creamy white, thin, and easily wrinkled. The corolla was fused basally into a narrow tube approximately 3–4 cm long and divided apically into three similar lobes, each measuring 2–3 cm long and 0.2–0.3 cm wide (Figure 2g). The two lateral staminodes were petaloid, approximately 1 cm long and 0.5 cm wide, creamy white, thin, and easily wrinkled. They tapered toward the base and broadened apically into petal-like structures with curled margins. The labellum was broadly petaloid, approximately 2 cm long and 1 cm wide, and creamy

white, with two purple stripes in the central region and two pale yellow markings in the lower portion (Figure 2h). Its distal portion expanded into a broad petal-like structure with curled margins. The fertile stamen was sessile or nearly sessile (Figure 2i). The anther was oval, approximately 0.5×0.2 cm, and pale yellow. Its membranous connective appendage was divided into two oval lobes approximately 0.2 cm long. The ovary was oval, approximately 0.4×0.2 cm, creamy white, smooth, and trilobular, with axile placentation and numerous small ovules in each locule (Figure 2j).

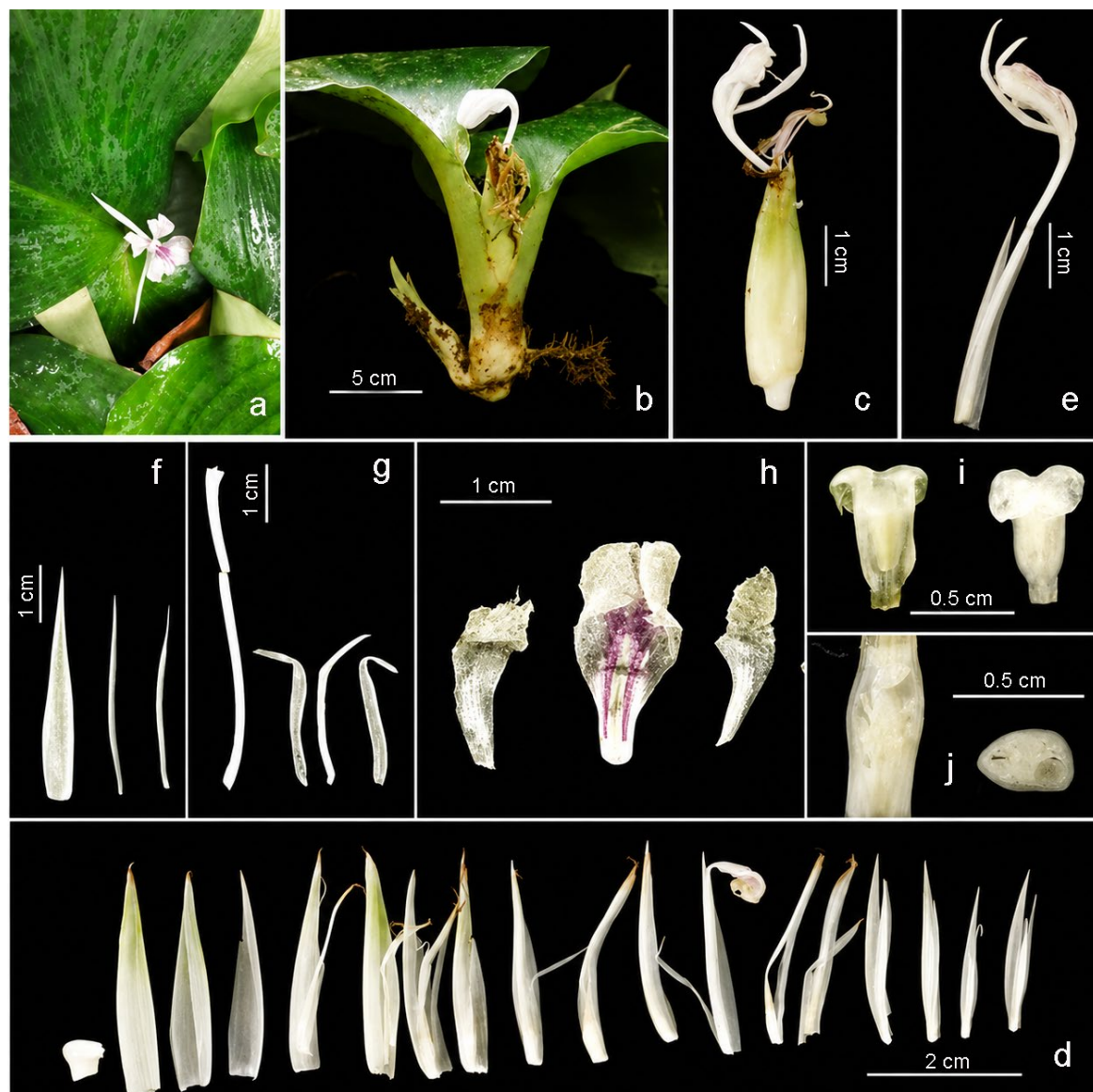


Figure 2. Macromorphological characteristics of the reproductive organs of *Kaempferia galanga* L.: (a) flowering plant in its natural habitat; (b) flowering plant; (c) intact inflorescence; (d) dissected inflorescence and its components; (e) complete flower; (f) bract and calyx; (g) corolla; (h) labellum and lateral staminodes; (i) fertile stamen; (j) ovary

3.2 Anatomical Characteristics

3.2.1 Leaf

The leaf exhibited an isobilateral monocotyledonous structure (Figure 3a). The adaxial surface was slightly concave, whereas the abaxial surface was V-shaped and convex. Both the upper and lower epidermises consisted of a single layer of unequal-sized elliptical cells. Stomata were present on both epidermal surfaces, although those on the adaxial surface appeared smaller than those on the abaxial surface (Figures 3b and 3d). The mesophyll consisted of thin-walled, rounded parenchymatous cells arranged irregularly, with cell size generally increasing toward the inner region. The vascular bundles were regularly distributed but varied in size, with alternating large

and small bundles forming a prominent arc near the lower epidermis and a second inner arc of smaller bundles. Each vascular bundle consisted of xylem positioned toward the adaxial surface and phloem located beneath it. Sclerenchyma tissue was present at both ends of the vascular bundles, forming an arc-like sheath around the xylem and phloem. The sclerenchyma adjacent to the phloem consisted of smaller, dark green-stained cells, whereas that adjacent to the xylem comprised larger cells with relatively thinner walls. Well-developed intercellular air spaces, or aerenchyma, occurred between the vascular bundles and were surrounded by parenchymatous cells containing numerous chloroplasts (Figure 3c).

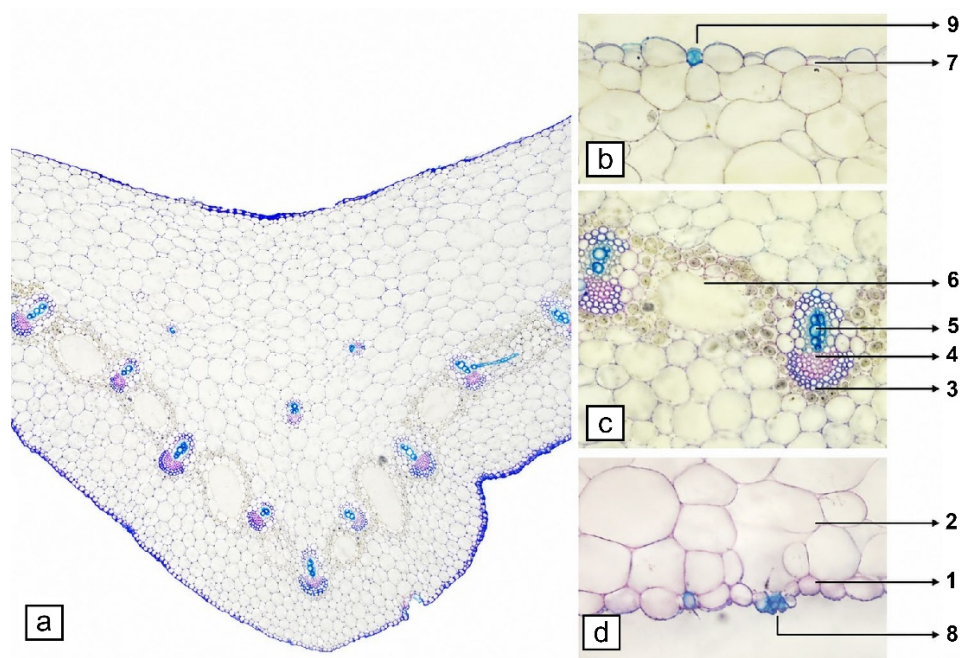


Figure 3. Anatomical characteristics of the leaf of *Kaempferia galanga* L.: (a) overall transverse section of the leaf; (b) enlarged view of the upper epidermis and stoma; (c) enlarged view of the mesophyll, vascular bundle, sclerenchyma, and aerenchyma; (d) enlarged view of the lower epidermis and stoma

Note: 1. lower epidermis; 2. mesophyll; 3. sclerenchyma; 4. phloem; 5. xylem; 6. aerenchyma; 7. upper epidermis; and 8–9. stomata.

3.2.2 Rhizome

The rhizome had an approximately circular cross-section (Figures 4a–4c). From the outermost to the innermost region, the cork consisted of 6–15 rows of rectangular cells arranged in radial and concentric patterns. Beneath the cork was the cortical parenchyma, which consisted of thin-walled, pink-stained cells of varying sizes and polygonal to slightly rounded shapes. Starch granules, scattered vascular bundles, and oil-containing cells were distributed within the cortical parenchyma (Figures 4d–4f). The starch granules were elliptical to nearly spherical and occurred either individually or in clusters. The vascular bundles were relatively small and consisted of xylem, comprising two to six green-stained cells, and phloem, composed of smaller dark pink-stained cells. The endodermis formed a distinct green-stained ring, with abundant starch granules located internally. The pericycle consisted of pink-stained cells adjacent to and alternating with the endodermal cells. Numerous vascular bundles occurred within the region enclosed by the pericycle. The pith parenchyma consisted of cells smaller than those of the cortical parenchyma and had slightly thicker cell walls. These cells contained abundant starch granules or oil droplets, and small vascular bundles were scattered throughout the pith (Figures 4e and 4f).

3.2.3 Root

The root had an approximately circular cross-section (Figure 5a). The cork consisted of 3–10 rows of rectangular cells arranged in radial files. Beneath the cork was a broad cortical region composed of unequal-sized, thin-walled, pink-stained cells that were polygonal to slightly rounded. The cortical parenchyma occupied a larger proportion of the root cross-section than the inner stele and contained scattered oil-containing cells (Figures 5b–5d). The endodermis consisted of a single layer of oval, thick-walled, green-stained cells. Immediately beneath it, the pericycle comprised pink-stained cells arranged alternately with the endodermal cells and surrounding the vascular tissues. The vascular bundles were small and concentrated near the pericycle. The primary xylem developed centripetally and contained one or two large vessels, whereas the primary phloem consisted of smaller dark red-stained cells located alternately between the xylem strands. The pith parenchyma was narrower than the cortical parenchyma and occasionally contained oil droplets (Figures 5c and 5d).

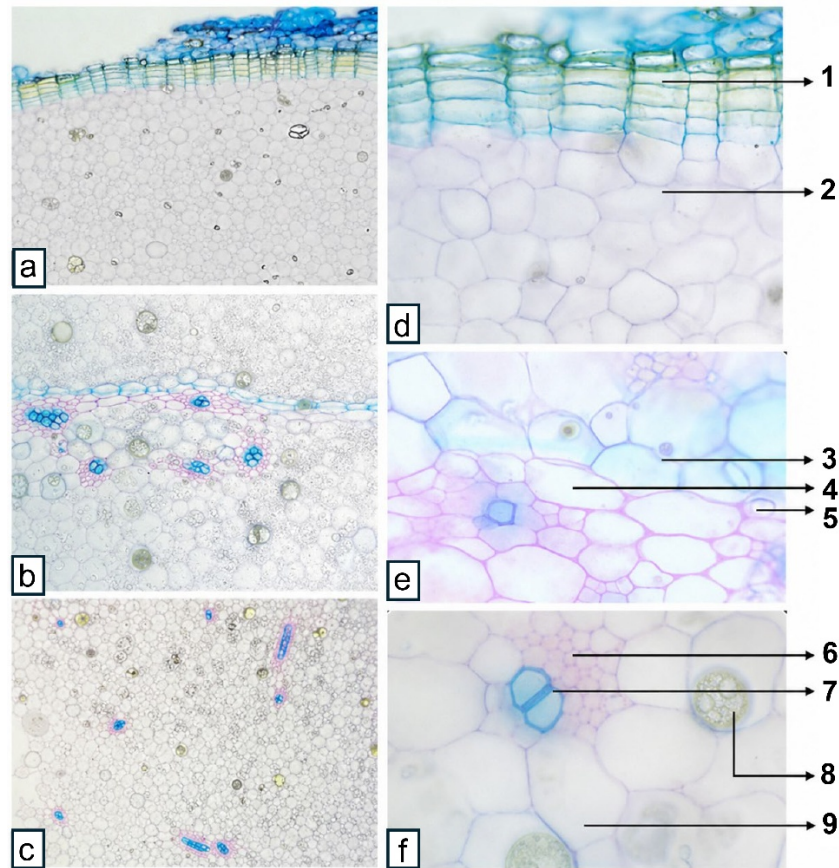


Figure 4. Anatomical characteristics of the rhizome of *Kaempferia galanga* L.: (a–c) transverse sections showing the rhizome tissues from the outer to the inner regions; (d–f) enlarged views of representative anatomical structures

Note: 1. cork; 2. cortical parenchyma; 3. endodermis; 4. pericycle; 5. starch granules; 6. phloem; 7. xylem; 8. oil-containing cells; and 9. pith parenchyma.

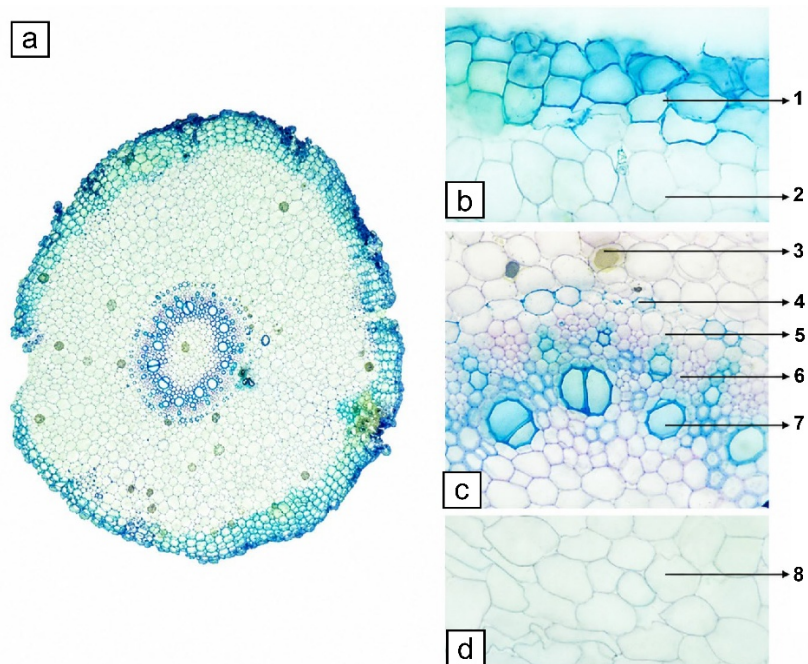


Figure 5. Anatomical characteristics of the root of *Kaempferia galanga* L.: (a) overall transverse section of the root; (b–d) enlarged views of representative anatomical structures

Note: 1. cork; 2. cortical parenchyma; 3. oil-containing cells; 4. endodermis; 5. pericycle; 6. phloem; 7. xylem; and 8. pith parenchyma.

3.3 Powder Microscopic Characteristics

3.3.1 Leaf powder characteristics

The leaf powder was dark green and had a faint aromatic odor. Microscopic examination revealed epidermal fragments bearing stomata measuring approximately $30 \times 50 \mu\text{m}$ (Figure 6a), together with fragments of parenchymatous tissue (Figure 6b) and spiral vessels (Figure 6c). The starch granules were oval to nearly spherical, possessed a small slit-like hilum, and measured approximately $20\text{--}30 \mu\text{m}$ in length (Figure 6d). Prismatic calcium oxalate crystals measuring approximately $20 \times 30 \mu\text{m}$ were also observed (Figure 6e). Multicellular covering trichomes with pointed apices and relatively thick walls measured approximately $150 \mu\text{m}$ in length, whereas longer trichomes reached up to $400 \mu\text{m}$ (Figures 6f and 6g).



Figure 6. Selected powder microscopic characteristics of the leaf of *Kaempferia galanga* L.: (a) epidermal fragment bearing a stoma; (b) parenchymatous tissue fragment; (c) spiral vessel; (d) starch granule; (e) prismatic calcium oxalate crystal; (f, g) multicellular covering trichomes

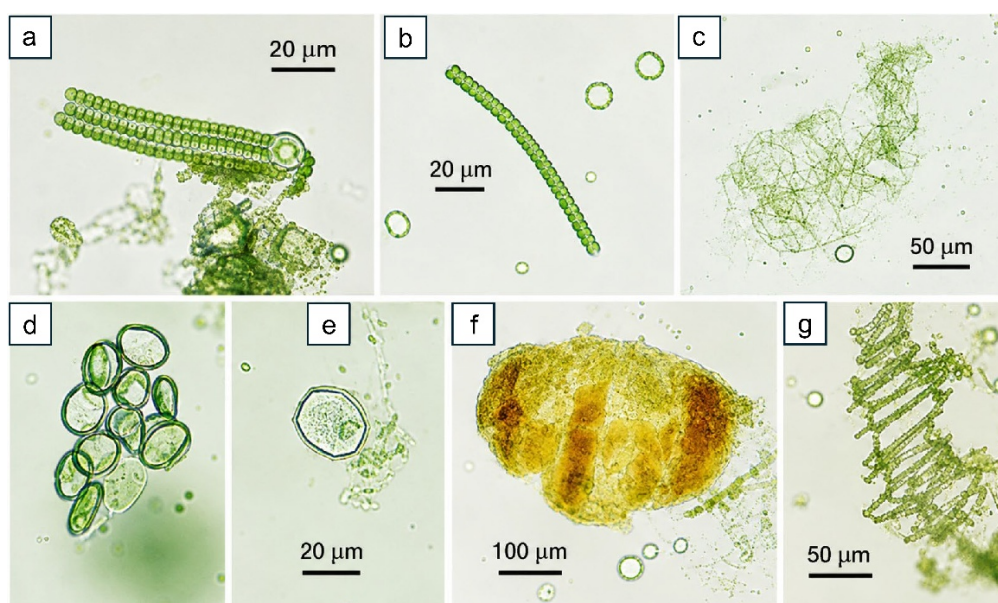


Figure 7. Selected powder microscopic characteristics of the rhizome of *Kaempferia galanga* L.: (a) spiral vessel fragment; (b) fiber; (c) parenchymatous tissue fragment; (d) cluster of starch granules; (e) individual starch granule; (f) pigmented mass; (g) cork fragment

3.3.2 Rhizome powder characteristics

The rhizome powder was creamy white and fine in texture, occasionally contained brown fragments, and had a characteristic aromatic odor. Microscopic examination revealed scattered spiral vessel fragments (Figure 7a), fibers (Figure 7b), and fragments of parenchymatous tissue (Figure 7c), some of which contained starch granules or pigmented substances. Numerous starch granules were observed. They were oval to nearly spherical, possessed a distinct hilum and faint concentric arcuate striations, and measured approximately 20–30 µm in length. The starch granules occurred either in clusters (Figure 7d) or individually (Figure 7e). Yellowish-brown pigmented masses were also observed (Figure 7f). Cork fragments consisted of thick-walled rectangular cells arranged in rows (Figure 7g).

3.4 Integrated Pharmacognostic Characteristics

The principal macromorphological, anatomical, and powder microscopic characteristics observed in the examined specimens are summarized in Table 1. Together, these characteristics constitute a preliminary integrated pharmacognostic profile of *K. galanga* collected from Bac Ninh Province, Vietnam. The profile combines external morphological features with internal anatomical structures and powder microscopic markers that may support botanical identification and quality control. However, the potential diagnostic relevance of these characteristics requires further validation using samples from broader geographical populations and closely related species.

Table 1. Summary of the pharmacognostic characteristics of *Kaempferia galanga* L. collected from Bac Ninh Province, Vietnam

Category	Diagnostic Characteristics	Potential Diagnostic Relevance
Macromorphology	Aromatic globose or chain-like rhizome; two broad ovate leaves; white flowers with a purple-striped labellum; 15–20 flowers per inflorescence	Preliminary field identification
Leaf anatomy	Isobilateral amphistomatic leaf with aerenchyma and scattered collateral vascular bundles	Anatomical authentication
Rhizome anatomy	Cork comprising 6–15 cell layers; abundant starch granules; oil-containing cells; scattered collateral vascular bundles	Authentication of medicinal rhizomes
Root anatomy	Cork comprising 3–10 cell layers; broad cortex with oil-containing cells; radial vascular tissues; narrow pith	Differentiation of underground organs
Leaf powder microscopy	Stomata; multicellular trichomes; spiral vessels; starch granules; calcium oxalate crystals	Authentication of powdered leaves
Rhizome powder microscopy	Numerous starch granules; cork fragments; spiral vessels; fibers; pigmented masses	Quality control of commercial rhizome powder
Integrated pharmacognostic profile	Combined macromorphological, anatomical, and powder microscopic characteristics	Support for integrated authentication and pharmacopoeial standard development

4. Discussion

4.1 Pharmacognostic Characterization and Diagnostic Value of *Kaempferia galanga* from Northern Vietnam

Morphological characteristics have long been regarded as an important basis for plant taxonomy and species identification because they reflect differences in plant form, structure, and evolutionary relationships (Viscosi & Cardini, 2011). In *K. galanga*, the morphology of the rhizomes, leaves, and flowers provides useful diagnostic characteristics for distinguishing the species from morphologically similar taxa within the genus *Kaempferia* (Gangmei et al., 2024; Saensouk & Saensouk, 2021). However, because macromorphological traits may vary with environmental conditions and developmental stages, the integration of anatomical and powder microscopic characteristics may improve the reliability of botanical authentication and medicinal-material quality control. Previous studies have reported the occurrence of *K. galanga* in Vietnam (Do et al., 2026). Although considerable attention has been paid to its phytochemical constituents and biological activities, detailed pharmacognostic descriptions, particularly of the anatomical characteristics of *K. galanga* materials collected in Vietnam, remain limited (Do & Dao, 2026; Men et al., 2020; Men et al., 2024; Men et al., 2022). Therefore, the present study provides baseline pharmacognostic information that may support the authentication and quality control of *K. galanga* collected from northern Vietnam.

The leaf anatomy observed in the present study displayed several characteristics previously reported for *K. galanga* and related *Kaempferia* species. Stomata occurred on both epidermal surfaces, consistent with the amphistomatic leaf structure reported by Paopun (2020). The leaves also exhibited an isobilateral structure,

scattered collateral vascular bundles embedded in the mesophyll, and well-developed aerenchyma, broadly comparable with previous foliar-anatomical descriptions of *K. galanga* and other members of the genus (Aishwarya, 2019; Das et al., 2004). The present study also documented stomata on both leaf surfaces and provided a detailed description of the sclerenchymatous tissue surrounding the vascular bundles and the distribution of aerenchyma. These minor differences from previous reports may be associated with environmental conditions, developmental stage, or geographical variation among populations and should be further investigated using specimens from additional populations and complementary anatomical or micromorphological techniques.

Similarly, the rhizomes exhibited characteristic anatomical features of the Zingiberaceae, including well-developed parenchymatous tissue, abundant starch granules, numerous oil-containing cells, and scattered collateral vascular bundles, in agreement with previous reports (Indriyani, 2017; Kumar, 2020). The present study additionally documented the number of cork cell layers, the structure of the endodermis and pericycle, and the differential staining patterns observed under the applied histological procedure. The roots displayed typical characteristics of the family Zingiberaceae, including a well-developed cork layer, broad cortical parenchyma, a distinct endodermis, radial vascular tissues, and oil-containing cells within the parenchyma (Saensouk & Saensouk, 2021). These observations were generally consistent with the descriptions reported by Kumar (2020) and Uma & Muthukumar (2014). The simultaneous characterization of the root and rhizome anatomy may assist in differentiating these underground organs during pharmacognostic examination and improve their diagnostic value for the quality control of *K. galanga* medicinal materials.

In pharmacognostic quality control, powder microscopy is considered an effective method for authenticating medicinal plant materials because many commercial herbal products are marketed in dried or powdered forms, in which external morphological characteristics are no longer preserved (Aeri et al., 2020; Ichim et al., 2020; World Health Organization, 2011). In the present study, the leaf powder was characterized by epidermal fragments bearing stomata, oval to nearly spherical starch granules, prismatic calcium oxalate crystals, spiral vessels, and multicellular covering trichomes. In contrast, the rhizome powder contained abundant starch granules occurring singly or in clusters, together with cork fragments, spiral vessels, fibers, and yellowish-brown pigmented masses. These characteristics may provide useful markers for the preliminary authentication and quality assessment of powdered *K. galanga* materials. Compared with other members of the family Zingiberaceae, such as *Alpinia galanga* and *A. calcarata*, differences in starch-granule morphology, crystal occurrence, and tissue fragments may contribute to species discrimination (Mathew et al., 2014).

The integration of macromorphological, anatomical, and powder microscopic characteristics provides a broader basis for authentication than external morphology alone. External morphological characteristics may vary according to environmental conditions, geographical distribution, and developmental stage, whereas several anatomical and microscopic characteristics are generally more stable and remain observable after plant materials have been dried or powdered. Characteristics such as leaf structure, vascular-bundle arrangement, oil-containing cells, starch granules, spiral vessels, trichomes, and calcium oxalate crystals are therefore relevant to the authentication, quality assessment, and standardization of medicinal plant materials (Edo et al., 2024; Ichim et al., 2020; World Health Organization, 2011). Nevertheless, the diagnostic characteristics documented in this study require validation using specimens from broader geographical populations, closely related species, and known adulterants before they can be adopted as standardized authentication criteria.

Several limitations should be acknowledged. The investigated specimens were collected from a single geographical location in northern Vietnam and may not represent the full variation among *K. galanga* populations throughout the country. In addition, the study did not include molecular authentication, phytochemical fingerprinting, closely related species, or commercial adulterants. Future studies should incorporate broader geographical sampling and complementary molecular, phytochemical, and chemometric approaches to validate the proposed diagnostic characteristics and assess their suitability for pharmacopoeial standard development.

4.2 Implications for Quality Assurance, Traceability, and Sustainable Medicinal Plant Production

The pharmacognostic characteristics documented in the present study have important implications not only for botanical identification but also for the quality control and sustainable utilization of medicinal plants in Vietnam. According to the World Health Organization (WHO), accurate authentication of medicinal plant species represents the first step in the implementation of GACP and constitutes a foundation for quality assurance throughout the medicinal plant production chain (Graf vom Hagen-Plettenberg et al., 2012; World Health Organization, 2003). Reliable species identification helps ensure that authentic planting materials are introduced into cultivation, thereby reducing the risk of misidentification and substitution and supporting greater consistency in medicinal raw materials (World Health Organization, 2003; World Health Organization, 2011).

In Vietnam, the cultivation of *K. galanga* has gradually expanded in response to increasing demand from the pharmaceutical industry and the traditional medicine and herbal-product sectors. Previous studies have demonstrated the considerable potential of this species, particularly the antioxidant, antimicrobial, and anti-inflammatory activities of its rhizomes and those associated with its major bioactive constituent, ethyl *p*-

methoxycinnamate (Do & Dao, 2026; Men et al., 2020; Men et al., 2024; Men et al., 2022; Umar et al., 2014). However, most studies have focused primarily on phytochemical composition and biological activities, whereas information on macromorphological, anatomical, and powder microscopic characteristics for species authentication and quality control remains limited. The close morphological similarity among species of the genus *Kaempferia*, together with the widespread commercialization of dried and powdered herbal materials, increases the risk of species misidentification and adulteration during production and trade (Ichim et al., 2020; Wongsuwan et al., 2025). The characteristics documented in the present study represent candidate diagnostic features that could potentially be used at different stages of the medicinal plant production chain, from germplasm selection and cultivation to harvesting, postharvest processing, and quality inspection. Consequently, these features may support raw-material traceability and the development of standardized quality-control procedures, subject to further validation.

Beyond quality control, accurate botanical authentication is relevant to sustainable medicinal plant production and production systems incorporating organic principles, because certified and traceable production systems require correctly identified planting materials. The diagnostic characteristics documented in this study may support the selection and propagation of authentic local germplasm, raw-material traceability, and the prevention of species substitution during cultivation, harvesting, and processing. They may also contribute to reducing the unintended collection or cultivation of closely related species. Once validated across broader geographical populations and against related taxa, the proposed pharmacognostic profile may contribute to the development of national quality standards and future pharmacopoeial monographs. These potential applications are consistent with broader objectives of sustainable and organic production systems, including quality assurance, certification, germplasm conservation, and responsible medicinal plant utilization.

5. Conclusions

This study characterized the macromorphological, anatomical, and powder microscopic characteristics of *K. galanga* collected from Bac Ninh Province, Vietnam. The macromorphological features, including the tuberous rhizome with a characteristic aromatic odor, leaves growing close to the ground, and white flowers with a purple-striped labellum, were consistent with typical descriptions of this species. Anatomical examination of the leaves, roots, and rhizomes revealed characteristic structural features of the family Zingiberaceae, including well-developed parenchymatous tissue, characteristic vascular-tissue arrangements, starch granules, and oil-containing cells. Microscopic analysis of the powdered leaf and rhizome materials identified potentially useful diagnostic features, including epidermal fragments bearing stomata, spiral vessels, calcium oxalate crystals, starch granules, sclerenchymatous fibers, and cork fragments.

These findings provide baseline pharmacognostic information for the identification and quality assessment of *K. galanga* from northern Vietnam. The diagnostic characteristics documented in this study may support the authentication of planting materials and commercial herbal products, thereby contributing to adulteration prevention, quality assurance, germplasm conservation, and the responsible cultivation and utilization of *K. galanga*. However, because this study was based on samples collected from a single geographical location and did not include closely related species or molecular and phytochemical validation, the proposed diagnostic characteristics should be interpreted as preliminary criteria.

Further studies incorporating broader geographical sampling, authenticated related species, and complementary molecular and phytochemical approaches are warranted to validate and extend these findings and to facilitate the development of pharmacopoeial standards. Once validated, these diagnostic characteristics may support the traceability and quality assurance of planting materials and commercial herbal products.

Author Contributions

Conceptualization, T.D.D. and M.H.L.; methodology, T.D.D. and T.T.L.B.; validation, T.D.D. and M.H.L.; formal analysis, T.D.D.; investigation, T.D.D. and T.T.L.B.; resources, M.H.L.; data curation, T.D.D.; writing—original draft preparation, T.D.D.; writing—review and editing, M.H.L. and T.T.L.B.; visualization, T.D.D.; supervision, M.H.L.; project administration, T.D.D. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with English language editing and to improve the clarity and readability of the text. The AI tool was not used to generate, analyze, or interpret research data, nor to create scientific conclusions or references. The authors carefully reviewed and edited all AI-assisted content and take full responsibility for the accuracy, originality, and integrity of the manuscript.

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